

GENERATED RESEARCH OUTPUT

Bitcoin Bridge Market Map

The mathematical map, institutional requirements, and companies occupying the bridge

Commercial territory forms where companies connect Bitcoin's verifiable state and signing rules to the institutional facts required by finance and law.

Generation contract

Every capability, equation, definition, bridge, company and evidence link on the following pages was read from the repository's structured domain model. Editing the model and rebuilding changes this document.

Model version: **0.1.0**

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1. The mathematical map

Bitcoin supplies narrow, machine-verifiable guarantees. The model requires every equation to define its terms and state both what it establishes and what it does not establish.

Capability and formula	Terms	Boundary
Signature validity $\text{Verify}(\text{pk}, \text{m}, \text{sig}) = 1$	pk = public key m = exact message or transaction sig = signature 1 = the verification rule accepts the signature	Establishes: The cryptographic verification rule accepted the signature for the supplied data. Does not establish: legal identity, legal ownership, lawful authority, voluntary intent
Threshold control $\text{valid signers} \geq t \text{ of } n$	valid signers = participants satisfying the signing policy t = minimum number required n = total configured participants $\geq$ = at least	Establishes: The configured cryptographic threshold was satisfied. Does not establish: organizational authority, fiduciary compliance
Ledger state $\text{Balance}(\text{U}, \text{h}) = \text{sum}(\text{value}(\text{u}))$	U = selected set of unspent outputs h = block height u = one unspent output sum = addition of the selected output values	Establishes: The selected unspent outputs existed in the accepted ledger state at the specified height. Does not establish: beneficial ownership, completeness of disclosure

2. What finance and law require

The bridge market exists because institutional capital needs facts and controls that are intentionally outside Bitcoin's consensus rules.

Traditional-finance requirements

Requirement	Stable model identifier
Client and beneficial-owner identity	identity
Account-level asset entitlement	asset-entitlement
Authenticated institutional authorization	authorization
Accounting, valuation, and reconciliation	accounting
Asset and liability coverage	solvency
Contractual and regulated recourse	recourse

Legal requirements

Requirement	Stable model identifier
Legal ownership or beneficial entitlement	legal-ownership
Capacity and legal personality	legal-capacity
Authority under mandate, office, trust, or court order	lawful-authority
Death, probate, trust, and beneficiary succession	succession
Enforceable remedies after fraud, error, or breach	remedies

Bitcoin evidence + institutional records = financeable Bitcoin

3. The bridge territory

Companies are grouped by the bridge classifications encoded in the model. A company may occupy more than one bridge.

Bridge	Bitcoin capabilities	Institutional requirements	Companies
Access and execution	ledger-state	accounting	Anchorage Digital , Coinbase Custody , Fidelity Digital Assets , Kraken , River
Regulated custody	signature-validity, threshold-control, ledger-state	identity, asset-entitlement, authorization, recourse, legal-ownership, legal-capacity, lawful-authority, remedies	Anchorage Digital , BitGo , Coinbase Custody , Copper , Fidelity Digital Assets
Collaborative custody	threshold-control	authorization, lawful-authority, succession	BitGo , Casa , Nunchuk , Unchained
Wallet governance	signature-validity, threshold-control	authorization, lawful-authority	BitGo , Copper , Fireblocks , Fordefi
Reserve and audit evidence	ledger-state, signature-validity	accounting, solvency, legal-ownership	Kraken , River
Blockchain attribution	ledger-state	identity, legal-capacity	Chainalysis , Elliptic , TRM Labs
Succession and continuity	threshold-control	identity, asset-entitlement, legal-ownership, lawful-authority, succession	Casa , Unchained

4. Evidence-backed claims

Claims are published with an explicit status and source set. Validation rejects unknown statuses and missing source references.

Status	Claim	Sources
CORROBORATED	A valid signature does not by itself establish legal ownership or lawful authority.	BIP-322 Generic Signed Message Format https://github.com/bitcoin/bips/blob/master/bip-0322.mediawiki Considerations for Auditing Cryptocurrencies https://www.icaew.com/technical/technology/blockchain-and-cryptoassets/blockchain-helpsheets/considerations-for-auditing-cryptocurrencies
CORROBORATED	Institutional crypto safekeeping remains subject to existing custody, fiduciary, information-security, and legal requirements.	Crypto-Asset Safekeeping by Banking Organizations https://www.occ.gov/news-issuances/news-releases/2025/nr-ia-2025-68a.pdf

Current scope

The model currently contains **3** Bitcoin capabilities, **6** finance requirements, **5** legal requirements, **7** bridge categories, **15** companies, **4** sources, and **2** evidence-linked claims.

This PDF demonstrates the engine, not a finished market census. Its visible gaps are model gaps that can now be improved through versioned contributions.